

BLOCK 5

EXTREME VALUES AND OPTIMISATION

Unit 16	Optimisation: An Introduction
Unit 17	Unconstrained and Constrained Optimisation
Unit 18	Advanced Topics in Optimisation– I
Unit 19	Advanced Topics in Optimisation – II



UNIT 16 OPTIMISATION: AN INTRODUCTION

Structure

- 16.0 Objectives
- 16.1 Introduction
- 16.2 Concept of Maxima and Minima
 - 16.2.1 Identification of Maxima and Minima
 - 16.2.2 Point of Inflexion
 - 16.2.3 A Conclusive Criterion
- 16.3 Extreme Values of Multivariate Functions
 - 16.3.1 Sufficient Condition for Extreme Values
 - 16.3.2 Function of Three Variables and Extrema
 - 16.3.3 Function of n Variables and Extrema
- 16.4 Let Us Sum Up
- 16.5 Key Words
- 16.6 Answers/Hints to Check Your Progress Exercises
- 16.7 Exercises

16.0 OBJECTIVES

After going through this unit, you should be able to:

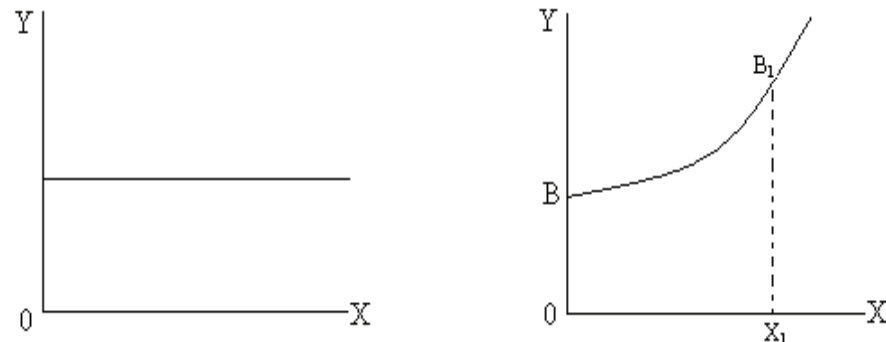
- define and identify the maximum and minimum points of a function;
- state the necessary conditions for maximum and minimum for functions;
- distinguish between a global and a local maximum/minimum;
- characterise an extremum; and
- distinguish an extremum from a point of inflexion.

16.1 INTRODUCTION

We have already learnt about the shapes of different functions and their curvature in Unit 4. To add on to our knowledge about functions, this unit provides us an understanding of the concepts like maximum and minimum of a given function. We will introduce the notion of a stationary point also where the first order derivatives of a function are zero and the function comes to a standstill momentarily. We then move on to the point of inflexion which is nothing but the point where the curve changes its curvature. We also learn about the extreme values of a function with several variables.

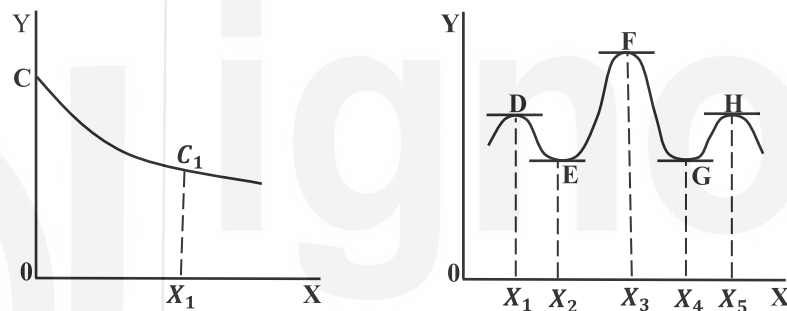
16.2 CONCEPT OF MAXIMA AND MINIMA

To start with, we assume the function $y = f(x)$ is differentiable (and hence continuous) throughout its range. The function may be represented by one of the following types of graphs:



i) Constant

ii) Monotonically increasing



iii) Monotonically decreasing

iv) Fluctuating

Fig. 16.1: Types of differentiable (hence continuous) functions

In the first case, the function is the same for all values of x . We shall consider the values of y as both maximum and minimum value.

In the second case, the value of the function is monotonically increasing. Let us restrict the domain of x to the interval $0 \leq x \leq x_1$. Then in this closed interval, y can assume a maximum of B_1x_1 and a minimum of OB . These maximum and minimum values of the function are called absolute maximum (global) and absolute (global) minimum respectively. Such maximum and minimum values of a function are called extreme values or extrema. Accordingly in the third case, OC becomes the global maxima and X_1C_1 the global minima if we restrict in the interval of x to $0 \leq x \leq x_1$.

Let us consider the fourth case. Here y oscillates with changes in x . The function generates three peaks, D , F and H and two bottoms E and G . The value of the function at D is the highest in comparison to the values at other points in its immediate neighborhood. Symbolically,

$$f(x_1 - \varepsilon) < f(x_1) \text{ and } f(x_1 + \varepsilon) > f(x_1) \text{ as } \varepsilon \rightarrow 0.$$

Thus, we can say that the value of the function at D is maxima, at least in some small interval $x_1 - \varepsilon < x < x_1 + \varepsilon$. Similarly, in another small interval $x_3 - \varepsilon < x < x_3 + \varepsilon$, the value of the function is maximum at $x = x_3$ and in the neighborhood of F,

$$f(x_3 - \varepsilon) < f(x_3) \text{ and } f(x_3 + \varepsilon) > f(x_3).$$

The function attains another maximum in the neighborhood of point F. These maximum values are called local (relative) maxima. A relative maximum gives us the information that the value of the function is maximum in the neighborhood of the maximum point only, and the function can assume higher values elsewhere. For example, in the fourth diagram of Fig 16.1, points D and H describe relative maximum, though the function assumes a higher value at F. The relative maxima are unlike global maxima. The global maximum is the highest value of the function with reference to its entire domain and hence unique.

A function can have several local maxima and unique global maxima. The concept of local minima and global minima can be explained in the same manner. For a continuous function, the global maximum must always be greater than the global minimum.

The local extrema always appear as point interior to an interval, however small. These local extrema are thus strictly interior extrema. In the following discussion, we shall consider only these types of extrema.

16.2.1 Identification of Maxima and Minima

Let us now look at maximum and minimum through Fig. 16.2.

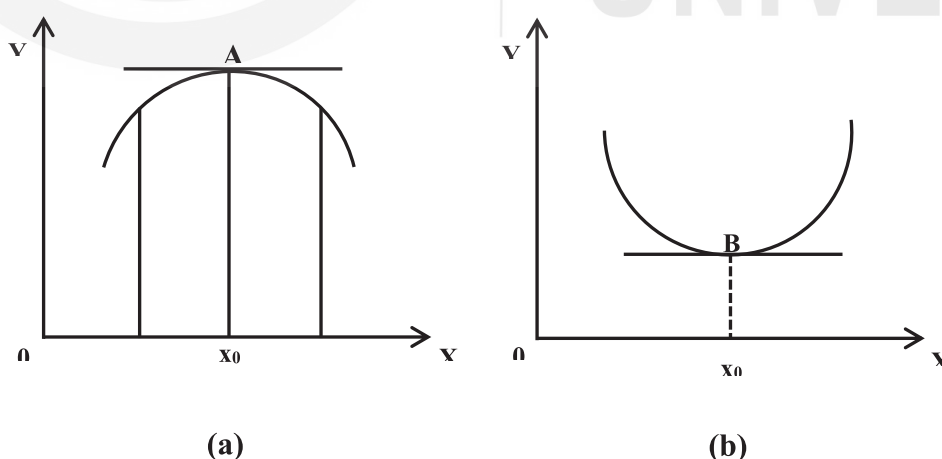


Fig. 16.2: Cases of Maximum and Minimum

The extremum described by panel (a) in Fig 16.2 is a maximum, while that of panel (b) is a minimum. In panel (a) the curve $y = f(x)$ takes a turn at point A. To the left of A, the curve is increasing i.e., $f'(x) > 0$ while to its right $f(x)$ is decreasing i.e., $f'(x) < 0$. Thus, at the turning point A, $f'(x)$ must be zero i.e., $f'(x_0) = 0$. Similarly, at point B in panel (b) of Fig. 16.2, $f'(x_0) = 0$. Therefore, we

can conclude that an extremum can take place only at the stationary points where $f'(x)=0$. The extreme values are always stationary values.

Let us construct ε in the neighborhood of the point x_0 (where x_0 is the point where the perpendicular from point A meets the x-axis) in panel (a). As A is local maximum, we know that in the interval $x_0 - \varepsilon < x < x_0 + \varepsilon$, the value of $f(x)$ at $x = x_0$ is the highest.

Thus, to the left of x_0 ,

$$x < x_0 \Rightarrow f(x) < f(x_0) \Rightarrow \frac{f(x) - f(x_0)}{x - x_0} > 0 \Rightarrow \lim_{x \rightarrow x_0^-} \frac{f(x) - f(x_0)}{x - x_0} > 0$$

Let the value of this limit be $l_1 \geq 0$. Then

$$\lim_{x \rightarrow x_0^-} \frac{f(x) - f(x_0)}{x - x_0} > l_1$$

To the right of x_0 ,

$$\begin{aligned} x > x_0 \Rightarrow f(x) < f(x_0) &\Rightarrow \frac{f(x) - f(x_0)}{x - x_0} < 0 \\ \Rightarrow \lim_{x \rightarrow x_0^+} \frac{f(x) - f(x_0)}{x - x_0} < 0 &= -l_2 \text{ (say) where, } l_2 \geq 0 \end{aligned}$$

Thus, we get the result that the right-hand limit and left-hand limit of x_0 assume two different values. Naturally, the limit does not exist. But the curve is smooth and continuous. So, the limit must exist at x_0 also. This can happen, if

$$\lim_{x \rightarrow x_0^-} \frac{f(x) - f(x_0)}{x - x_0} = \lim_{x \rightarrow x_0^+} \frac{f(x) - f(x_0)}{x - x_0}$$

$$l_1 = -l_2 \text{ i.e., } l_1 + l_2 = 0. \text{ i.e., if } l_1 = l_2 = 0.$$

The value of the first derivative at that point is zero is the necessary condition for a function to be maximum or minimum at that point.

We assume the function to be differentiable everywhere in the domain of its definition. To obtain sufficient condition for maxima and minima, we observe the pattern of change of the slope of the curves in panel (a) of Fig. 16.2. We saw that to the left of point A, $f'(x) > 0$ and to the right of A, $f'(x) < 0$. This means that while passing through the maximum point A, the slope of the curve $f'(x)$ changes its sign from positive to negative. This is the sufficient condition for maximum. This condition, along with the necessary condition, gives us the necessary and sufficient condition for an extremum to be maximum. Mathematically, the sufficient condition means $f''(x_0) \neq 0$. We find that the above two definitions are not equivalent. But we shall assume $f''(x_0)$ assumes a non-zero value.

Let us now summarise the result we have obtained so far.

i) *Necessary condition*

a) for maxima: $f''(x)=0$

b) for minima: $f'(x)=0$

ii) *Sufficient condition*

a) for maxima: $f'(x)=0$ and $f''(x)<0$

b) for minima: $f'(x)=0$ and $f''(x)>0$.

Example 16.1: Find the maxima and minima for the following function:

a) $y = 3x^4 - 10x^3 + 6x^2 + 5$

First order condition (F.O.C.):

$$12x^3 - 30x^2 + 12x = 0$$

$$\text{or, } 3x(4x - 2)(x - 2) = 0.$$

Either, $x = 0$ or, $x = 2$ or $x = \frac{1}{2}$.

Second, order condition (S.O.C.):

$$36x^2 - 60x + 12 = 0$$

$$\text{At } x = 0, f''(x) = 36(0) - 60(0) + 12 \\ = 12 > 0.$$

$$\text{At } x = 2, f''(x) = 36(2)^2 - 60(2) + 12 \\ = 144 - 120 + 12 = 36 \\ = 36 > 0.$$

$$\text{At } x = \frac{1}{2}, f''(x) = 36\left(\frac{1}{2}\right)^2 - 60\left(\frac{1}{2}\right) + 12 \\ = 9 - 30 + 12 = -9 \\ = -9 < 0.$$

Hence, the function attains maximum at $x = \frac{1}{2}$ and minimum at $x=0$ and $x = 2$.

Example 16.2:

b) Show that for the function $y = x + \frac{1}{x}$, the maximum value is less than its minimum value.

$$f'(x) = 1 - \frac{1}{x^2}, \quad f''(x) = \frac{2}{x^3}$$

$$f'(x) = 0 \Rightarrow x = 1 \text{ or } -1$$

$$f''(1) = 2 > 0 \Rightarrow f(x) \text{ has its minima at } x = 1 \text{ and } f(1) = 2$$

$$f''(-1) = -2 < 0 \Rightarrow f(x) \text{ has its maxima at } x = -1 \text{ and } f(-1) = -2$$

Thus, the maximum value is less than its minimum value.

Check Your Progress 1

- 1) Show that $f(x) = x^3 - 6x^2 + 24x + 4$ has neither a maximum nor a minimum.

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- 2) Examine whether $x^{1/x}$ has a maximum or minimum point.

.....

16.2.2 Point of Inflexion

Now we will consider the cases for which the second derivative is zero. If the second derivative is zero at a point, the function may have a maximum, a minimum or a point of inflexion. The point of inflexion is defined as a point at which a curve changes its curvature.

At the point of inflexion, a curve does not change its direction of movement but only the nature of its slope (Fig 16.3). At each inflexion point, the first order derivative of the curve either reaches a maximum or a minimum. The sufficient condition for a point of inflexion is

$$f''(x) = 0 \text{ and } f'''(x) \neq 0$$

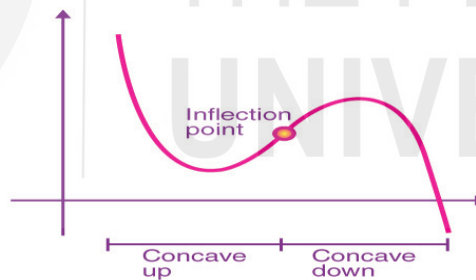


Fig.16.3: Inflexion point

Thus, if a function has $f'(x) = 0$, $f''(x) = 0$ and $f'''(x) \neq 0$ at the point x , the point is said to be stationary and inflexional.

At this stage, we state an important result:

If a monotonic function is first convex and then concave, the point of inflexion generates a maximum $f'(x)$. Similarly, if the function is initially concave, and then convex, the point of inflexion generates a minimum $f'(x)$.

Example 16.3: Find the point of inflexion for the function $f(x) = x^3 + 2$

Here, $f'(x) = 3x^2$, $f''(x) = 6x$ and $f'''(x) = 6 \neq 0$

When $f''(x) = 6x = 0$, $x = 0$. Thus, the function has point of inflexion at the origin.

Check Your Progress 2

- 1) Find the point of inflexion for the function $f(x) = x^3 + x^2 + x + 1$

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.....
.....
.....

16.2.3 A Conclusive Criterion

Let us expand the function $f(x)$ around the point $x = a$.

$$f(x) - f(a) = f'(a)(x - a) + \frac{f''(a)}{2!}(x - a)^2 + \dots + \frac{f^n(x^*)}{n!}(x - a)^n$$

where, $x^* = a + \theta(x - a)$, $0 \leq \theta \leq 1$.

The above series is the Taylor series with Lagrange's form of the remainder.

From this result we will consider various possible cases:

Case-I: $f'(a) \neq 0$

For simplicity, we assume $n = 0$. By Taylor's expansion,

$$f(x) - f(a) = f'(x^*)(x - a).$$

Since $f'(a) \neq 0$, $f'(x^*) \neq 0$. To the left of x^* , $(x - a) < 0$ and to its right, $(x - a) > 0$. Thus, to the left and the right of $x = a$, $f(x) - f(a)$ has opposite signs. This implies the extrema of a function is characterized by $f'(x) = 0$.

Case-II: $f'(a) = 0$, $f''(a) \neq 0$

Suppose $n = 1$. From the Taylor series,

$$f(x) - f(a) = f'(a)(x - a) + \frac{f''(x^*)}{2!}(x - a)^2 = \frac{1}{2} f''(x^*)(x - a)^2$$

Now, $(x - a)^2 > 0$ implies that the sign of $f(x) - f(a)$ depends on the sign of $f''(x^*)$ which again depends on the sign of $f''(a)$.

Thus if

- i) $f''(a) > 0$, $f(x) - f(a) > 0$ and the curve is minimum at $x = a$
- ii) $f''(a) < 0$, $f(x) - f(a) < 0$ and the curve is maximum at $x = a$

Case-III:

$$f'(a) = 0, f''(a) = 0 \text{ and } f'''(a) \neq 0$$

Let us assume $n=2$.

Here from the Taylor expansion,

$$f(x) - f(a) = f'(a)(x - a) + \frac{f''(a)}{2!}(x - a)^2 + \frac{f'''(x^*)}{3!}(x - a)^3 = \frac{1}{6} f'''(x^*)(x - a)^3$$

Here again $f'''(x^*) \neq 0$ and depends on the sign of $f'''(a)$. We find that $(x-a)^3 > 0$ to the right of a and < 0 to the left of a . Thus, the whole right-hand expression has opposite signs at the two opposite sides of a . Thus a is not an extremum, rather a point of inflexion.

Case-IV:

$$f'(a) = 0, f''(a) = 0, f'''(a) = 0$$

and

$$f^{(4)}(a) \neq 0$$

Here we assume $n=4$. Again, by Taylor expansion,

$$f(x) - f(a) = \frac{1}{24} f^{(4)}(x^*)(x-a)^4.$$

Since $(x-a)^4 > 0$, the sign of $f(x) - f(a)$ depends on the sign of $f^{(4)}(x^*)$ which in turn depends on the sign of $f^{(4)}(a)$.

We can conclude that

- i) $f^{(4)}(a) > 0$, $f(x) - f(a) > 0$ and the curve is minimum at $x = a$
- ii) $f^{(4)}(a) < 0$, $f(x) - f(a) < 0$ and the curve is maximum at $x = a$

Generalizing the above argument we may conclude;

When $f'(a) = 0, f''(a) = 0, \dots, f^{(n-1)}(a) = 0$ & $f^{(n)}(a) \neq 0$, the point $x = a$ has

- i) a maximum if n is even and $f^{(n)}(a) < 0$
- ii) a minimum if n is even and $f^{(n)}(a) > 0$
- iii) an inflexion if n is odd.

16.3 EXTREME VALUES OF MULTIVARIATE FUNCTIONS

Let $y = f(x)$ and the first order condition for optimisation is $f'(x) = 0$.

Now, $dy = f'(x)$.

Thus, F.O.C. for optimisation requires $dy = 0$.

To find out the extrema of a bivariate function, we proceed according to the same argument.

Consider the bivariate function $y = f(x_1, x_2)$.

$$dy = f_1 dx_1 + f_2 dx_2.$$

The first order condition for optimisation is

$$dy = f_1 dx_1 + f_2 dx_2 = 0$$

As both dx_1 and dx_2 are not equal to zero, the above equation can hold only when $f_1 = f_2 = 0$.

Hence, F.O.C. for optimum of a bivariate function are

$$f_1 = 0 \text{ and } f_2 = 0.$$